Experiment-4

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Approach

① We will check for 0 to 31st bit as no. are in integer range.
② For i^{th} bit we will count the no. of elements which has their ith bit set & unset, then sum += curBitSum(setCount * unsetCount).
③ We will do this for all 32 bit
④ Multiply by 2 and then provide the modulus by 10^7

For ex:-

A = [1, 3, 5]

Binary representation :-

```
  001
  011
  101
```

sum = 0
at i = 0

setCount = 3, unsetCount = 0

setCount * unsetCount = 0

sum = 0

at i = 1

setCount = 1, unsetCount = 2

sum += 2 = sum = 2

at i = 2

setCount = 1, unsetCount = 2

sum += 2 = sum = 4

Function for my approach :-

```
public int distance( int[] nums){
    int n = nums.length;
    int total =0;
    for(int bit =0; bit<32 ; bit ++){
        int ones =0;
        for(int num : nums){
            if((num & (1<< bit)) != 0){
                ones++;
            }
        }
        int zeros = n - ones;
        total += ones* zeros;
    }
    return total;
}
```